## Proposition 4.1, Subcase 2b

**Setting.** The quadratic model is

$$m_k(p) = f_k + g_k^\top p + \tfrac{1}{2} p^\top B_k p,$$

with $g_k \neq 0$, and the Cauchy point is $p_k^C = -\alpha_k^C g_k$, where $\alpha_k^C$ minimizes $m_k(-\alpha g_k)$ over $0 \leq \alpha \leq \Delta_k / \|g_k\|$. Define

$$\varphi(\alpha) := m_k(-\alpha g_k) = f_k - \alpha \|g_k\|^2 + \tfrac{1}{2} \alpha^2 \, g_k^\top B_k g_k.$$

Case 2 is $g_k^\top B_k g_k > 0$; then $\varphi$ is a strictly convex parabola with unconstrained minimizer

$$\alpha^* = \frac{\|g_k\|^2}{g_k^\top B_k g_k} > 0.$$

**Claim (Subcase 2b).** *If $g_k^\top B_k g_k > 0$ and $\alpha^* > \Delta_k / \|g_k\|$, then $\alpha_k^C = \Delta_k / \|g_k\|$ and*

$$m_k(0) - m_k(p_k^C) \;\geq\; \tfrac{1}{2} \|g_k\| \, \Delta_k.$$

*Proof.* **Step 1: $\alpha_k^C = \Delta_k / \|g_k\|$.** Since $\varphi'(\alpha) = -\|g_k\|^2 + \alpha \, g_k^\top B_k g_k$ and $g_k^\top B_k g_k > 0$, we have $\varphi'(\alpha) < 0$ for every $\alpha < \alpha^*$. Hence $\varphi$ is strictly decreasing on $[0, \alpha^*]$. The feasible interval $[0, \Delta_k / \|g_k\|]$ is contained in $[0, \alpha^*]$ (by hypothesis $\Delta_k / \|g_k\| < \alpha^*$), so the minimum of $\varphi$ over it is attained at the right endpoint: $\alpha_k^C = \Delta_k / \|g_k\|$.

**Step 2: bound on the decrease.** For any $\alpha \in [0, \alpha^*]$,

$$\varphi(0) - \varphi(\alpha) = \alpha \|g_k\|^2 - \tfrac{1}{2} \alpha^2 \, g_k^\top B_k g_k.$$

Since $0 \leq \alpha \leq \alpha^*$ and $g_k^\top B_k g_k > 0$,

$$\tfrac{1}{2} \alpha^2 \, g_k^\top B_k g_k \;\leq\; \tfrac{1}{2} \alpha \, \alpha^* \, g_k^\top B_k g_k = \tfrac{1}{2} \alpha \|g_k\|^2,$$

using $\alpha^* g_k^\top B_k g_k = \|g_k\|^2$. Therefore

$$\varphi(0) - \varphi(\alpha) \;\geq\; \alpha \|g_k\|^2 - \tfrac{1}{2} \alpha \|g_k\|^2 = \tfrac{1}{2} \alpha \|g_k\|^2. \tag{$*$}$$

**Step 3: conclusion.** Applying $(*)$ with $\alpha = \alpha_k^C = \Delta_k / \|g_k\| \in [0, \alpha^*]$, and noting that $\varphi(0) = m_k(0)$ and $\varphi(\alpha_k^C) = m_k(p_k^C)$,

$$m_k(0) - m_k(p_k^C) \;\geq\; \tfrac{1}{2} \cdot \frac{\Delta_k}{\|g_k\|} \cdot \|g_k\|^2 = \tfrac{1}{2} \|g_k\| \, \Delta_k.$$

Finally, since $\Delta_k \geq \min(\Delta_k, \|g_k\| / \|B_k\|)$, this yields inequality (13) in this subcase. □

**Remark** (Meaning of "the slope dominates"). *Inequality $(*)$ says that, as long as $\alpha$ does not exceed $\alpha^*$, the actual decrease is at least half of the linear decrease predicted by the slope at the origin, $-\varphi'(0) \, \alpha = \alpha \|g_k\|^2$. Indeed, the ratio of the curvature term to the linear term is*

$$\frac{\tfrac{1}{2} \alpha^2 \, g_k^\top B_k g_k}{\alpha \|g_k\|^2} = \frac{\alpha}{2\alpha^*} \;\leq\; \frac{1}{2} \qquad (0 < \alpha \leq \alpha^*).$$